Practical 3

To study and implement **Lists, Tuples, and Dictionaries** and perform various operations such as accessing, traversing, updating, deleting elements, and using built-in operators and methods.

Program 1: Create and Access List Elements

```
# Creating a List

fruits = ["Apple", "Banana", "Mango", "Orange"]

print("Complete List:", fruits)
print("First Element:", fruits[0])
print("Last Element:", fruits[-1])
```

Output

```
Complete List: ['Apple', 'Banana', 'Mango', 'Orange']
First Element: Apple
Last Element: Orange
```

Program 2: Traverse a List

```
numbers = [10,20,30,40,50]

print("Elements are:")

for item in numbers:
    print(item)
```

Program 3: Update and Delete List Elements

```
students = ["Ram","Shyam","Rita","Amit"]

print("Original List")
print(students)

students[1] = "Suresh"
```

```
print("After Update")
print(students)

del students[2]

print("After Delete")
print(students)
```

Program 4: List Methods

```
numbers = [5,2,8,1]

numbers.append(10)
numbers.insert(2,100)
numbers.remove(8)
numbers.sort()

print(numbers)
```

Program 5: List Operators

```
list1 = [1,2,3]
list2 = [4,5]

print("Concatenation:", list1 + list2)
print("Repetition:", list1 * 3)

print(2 in list1)
print(10 in list1)
```

Program 6: Create and Access Tuple

```
student = ("Rahul",20,"Mumbai")

print(student)

print("Name:", student[0])
print("Age:", student[1])
print("City:", student[2])
```

Program 7: Tuple Operators

```
t1 = (10,20,30)
t2 = (40,50)

print("Concatenation:", t1+t2)

print("Repetition:", t1*2)
```

```
print(20 in t1)

print(len(t1))
```

Program 8: Traverse Tuple

```
colors = ("Red","Green","Blue")

for color in colors:
    print(color)
```

Program 9: Dictionary Creation, Access and Update

```
student = {
    "Roll":101,
    "Name":"Amit",
    "Marks":88
}

print(student)

print(student["Name"])

student["Marks"] = 95

print(student)
```

Program 10: Dictionary Methods

```
student = {
    "Roll":101,
    "Name":"Amit",
    "Marks":90
}

student["City"] = "Mumbai"

print(student)

del student["Roll"]

print(student)

print(student.keys())

print(student.values())

print(student.items())
```

Program 11: Traverse Dictionary

```
student = {
    "Name":"Rahul",
    "Age":20,
    "City":"Mumbai"
}

for key,value in student.items():
    print(key,"=",value)
```

Program 12: Membership Operator

```
student = {
    "Name":"Rahul",
    "Age":20
}

print("Name" in student)
print("Marks" in student)
```

Important List Methods

Method	Description
append()	Add element at end
insert()	Insert element
remove()	Remove element
pop()	Delete element
sort()	Sort list
reverse()	Reverse list
clear()	Remove all elements

Important Dictionary Methods

Method	Description
keys()	Returns all keys

values()	Returns all values
items()	Returns key-value pairs
get()	Get value using key
update()	Update dictionary
pop()	Remove element
clear()	Delete all items

Operators

Operator	Example
Concatenation	`+`
Repetition	`*`
Membership	`in`, `not in`
